

Course Introduction

SEAPOWER THROUGH ENGINEERING

TOPIC LEARNING OBJECTIVES

Upon successful completion of this topic, the student will be able to:

1. Recognize the diverse backgrounds of classmates based on their career biographical introductions.
2. Relate Engineering Duty Officer (EDO) responsibilities to Navy priorities.
3. Identify major EDO competencies and sub-specialties.
4. Recognize what distinguishes EDOs from other Navy officers.
5. Recognize the purpose of the EDO Basic Course.
6. Identify major topics in the curricula of the Basic and Reserve courses.
7. Recognize the purpose of the EDO Senior Course.
8. Identify major topics in the EDO Senior Course Curriculum.
9. Given the Commanding Officer's expectations for and perspective on the Basic Course, identify student responsibilities.

STUDENT PREPARATION

Student Support Material

1. 2026 National Defense Strategy
   https://media.defense.gov/2026/Jan/23/2003864773/-1/-1/0/2026-NATIONAL-DEFENSE-STRATEGY.PDF
2. CNO 34: The Navy Campaign Plan (2025)
3. CNO34 Day One Message - Foundry - Fleet – Fight
   https://www.navy.mil/Leadership/Chief-of-Naval-Operations/display-cnotes/Article/4292705/cno34-day-one-message-foundry-fleet-fight/
4. C-NOte(s): Read and familiarize

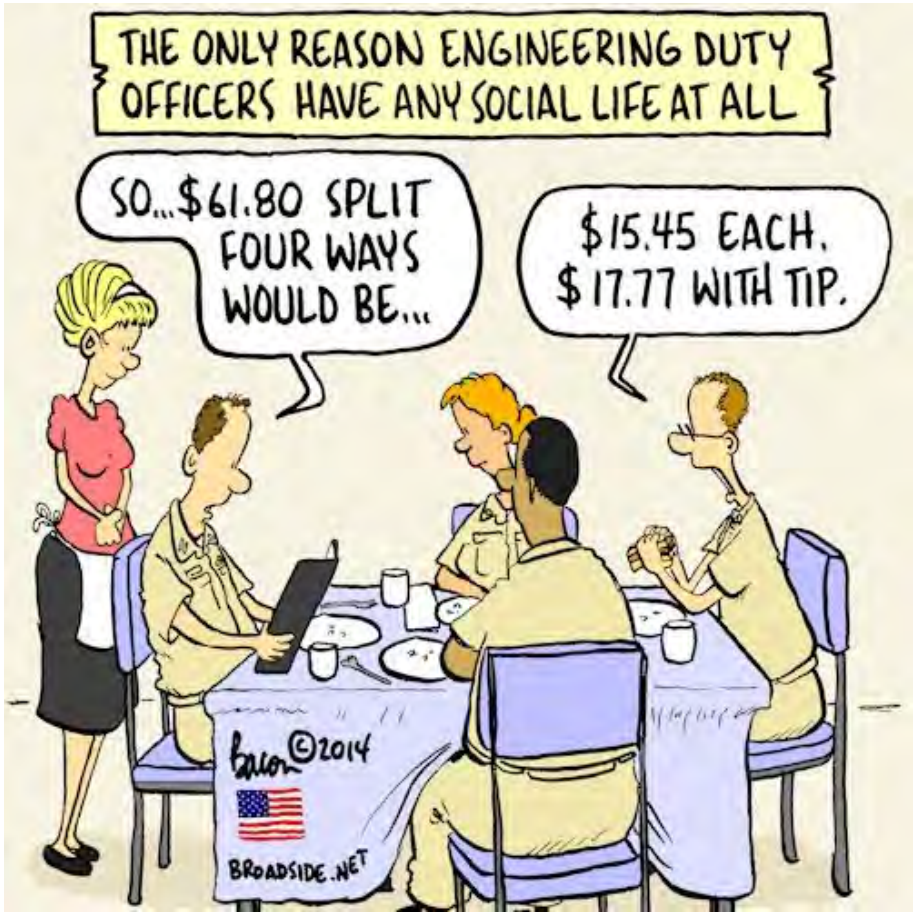
Primary References

1. Responsibilities and Procedures Concerning the Engineering Duty Officer School (NAVSEAINST 5400.56F)
2. EDO School Note "Non-Attribution Policy"

Additional References

1. "Engineering Duty…A Career Perspective." NAVPERS 15310, December 1978
2. "Evolution of the Engineering Duty Officer in the United States Navy." CAPT Jerome J. Fee, USN (Ret.). Naval Engineers Journal, Winter 2001
3. National Security Strategy (November 2025)
   https://www.whitehouse.gov/wp-content/uploads/2025/12/2025-National-Security-Strategy.pdf

EDOs as seen from the outside looking in…





Why Engineering Duty Officers usually drink alone

Welcome to Your School…and to the EDO Community!



Welcome

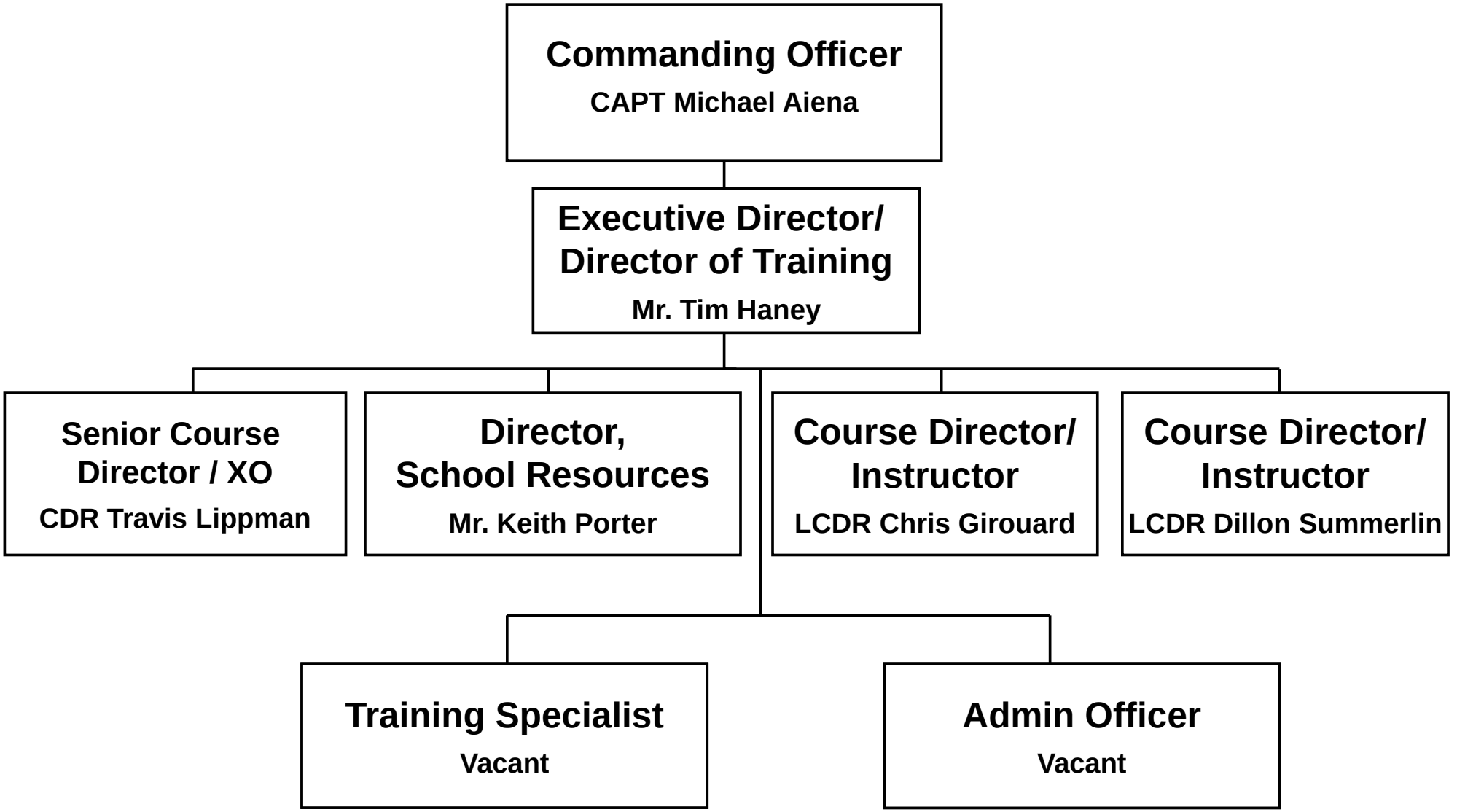
Naval Base Ventura County (NBVC)

Naval Education and Training Command (NETC)

Overview

- EDO School Organization
- Student Introductions
- EDO Community
- EDO School Courses
- CO Expectations

Engineering Duty Officer School Staff



CAPT Michael P. Aiena

- Born in New Orleans – GEAUX TIGERS!
- BS in History (USNA), MS in Applied Physics, (NPS)
- Surface Warfare Officer - USS ROBERT G. BRADLEY (FFG 49), USS LEYTE GULF (CG 55)
- Engineering Duty Officer - CSSQT Project Officer, NSWC Port Hueneme (Qual Tour)
- Assistant Program Manager (APM) for Mission Operations, PEO Space Systems, Navy Communications Satellite Program Office (PMW 146)
- US National Deputy for NATO SEASPARROW Project Office and Principal Assistant Program Manager for Surface EW and EO/IR, (PEO IWS 12, 2)
- Commanding Officer: NSWC Dahlgren Division Dam Neck Activity, NSWC Corona Division





CAPT Michael Aiena

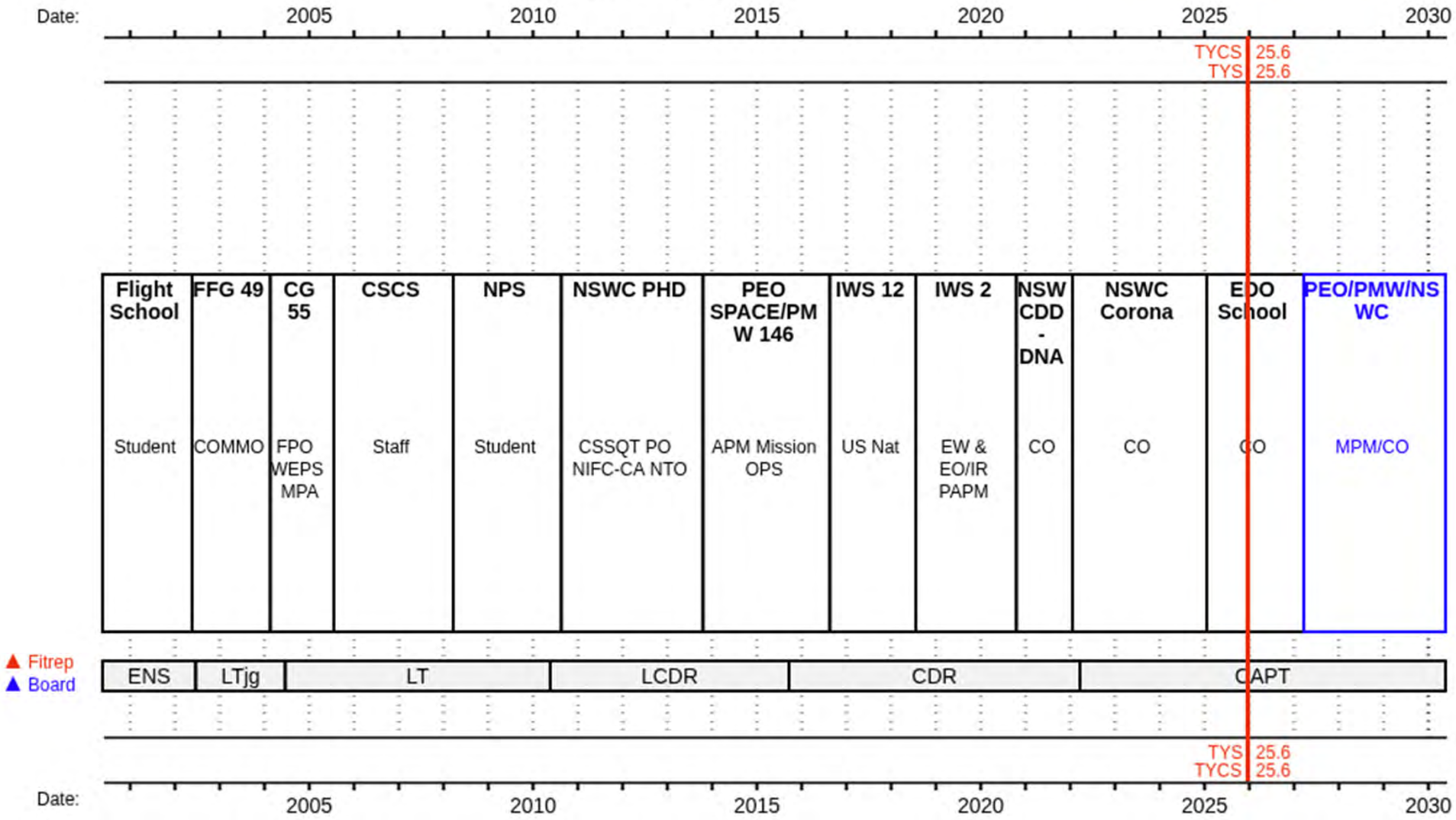
CO, EDO School

Career Goal: Major Program Manager, Combat Systems

Warfare Qual: Surface Warfare

Primary Mentor Group: CC
Secondary Mentor Group: IWE

DAWIA Certs: AA4 AN2 APM



Last Modified: 28-Feb-2025

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Introductions

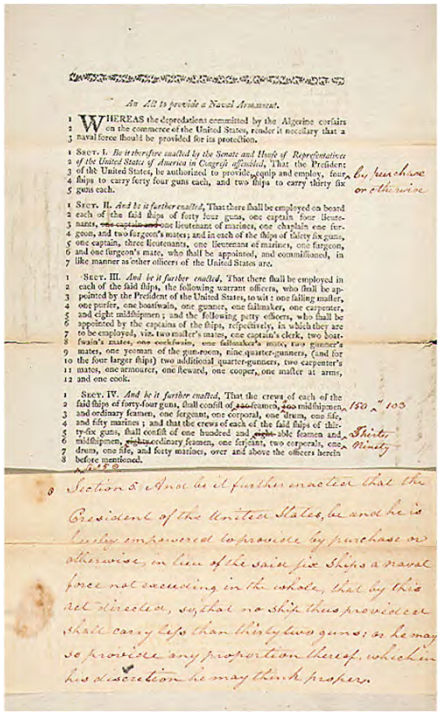
- Tent cards
  - Using the marker, write your preferred name in LARGE FONT
- Around the room (< 1 min per student)
  - Brief intro:
    - What is your preferred name?
    - What are your hobbies?
    - What would you like to learn about while here?

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Establishing a Navy

- 1794 – U.S. House appoints a committee to prepare a report on what kind of naval force would be necessary to defend against the current threat (Pirates)
  - 20 Jan – Committee reported resolution to authorize the procurement of six Frigates
  - Act to provide naval armament (Naval Act of 1794)
    - Department of War recommends the construction of new frigates designed to be superior to any vessel of that class in European Navies
    - Acquisition Guidance (Some of the first)
      - Minimize Labor cost by using government employees
      - Distribute construction sites geographically to spread economic benefit



An Act to provide a Naval Armament.

WHEREAS the depredations committed by the Algerine corsairs on the commerce of the United States, render it necessary that a naval force should be provided for its protection.

SECT. I. Be it therefore enacted by the Senate and House of Representatives of the United States of America in Congress assembled, That the President of the United States, be authorized to provide, equip and employ, four ships to carry forty four guns each, and two ships to carry thirty six guns each.

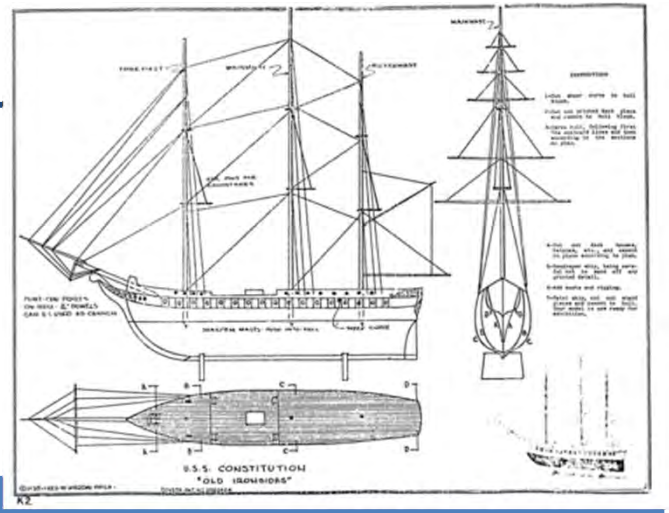
SECT. II. And be it further enacted, That there shall be employed on board each of the said ships of forty four guns, one captain four lieutenants, ~~one captain and~~ one lieutenant of marines, one chaplain one surgeon, and two surgeon's mates; and in each of the ships of thirty six guns, one captain, three lieutenants, one lieutenant of marines, one surgeon, and one surgeon's mate, who shall be appointed, and commissioned, in like manner as other officers of the United States are.

SECT. III. And be it further enacted, That there shall be employed in each of the said ships, the following warrant officers, who shall be appointed by the President of the United States, to wit: one sailing master, one purser, one boatswain, one gunner, one sailmaker, one carpenter, and eight midshipmen; and the following petty officers, who shall be appointed by the captains of the ships, respectively, in which they are to be employed, viz. two master's mates, one captain's clerk, two boatswain's mates, one cockswain, one sailmaker's mate, two gunner's mates, one yeoman of the gun-room, nine quarter-gunners, (and for the four larger ships) two additional quarter-gunners, two carpenter's mates, one armourer, one steward, one cooper, one master at arms, and one cook.

SECT. IV. And be it further enacted, That the crews of each of the said ships of forty-four guns, shall consist of ~~two hundred~~ seamen, ~~two~~ midshipmen, and ordinary seamen, one sergeant, one corporal, one drum, one fife, and fifty marines; and that the crews of each of the said ships of thirty-six guns, shall consist of one hundred and ~~eighty~~ able seamen and midshipmen, ~~eighty~~ ordinary seamen, one serjeant, two corporals, one drum, one fife, and forty marines, over and above the officers herein before mentioned.

Section 5. And be it further enacted, that the President of the United States, be and he is hereby empowered to provide by purchase or otherwise, in lieu of the said six ships a naval force not exceeding in the whole, that by this act directed, so that no ship thus provided for shall carry less than thirty two guns; as he may so provide any proportion thereof, which to his discretion he may think proper.

1st EDO Jobs



- To support the six construction sites
  - Naval Constructor hired at each site
  - Navy Captains appointed as superintendents

U.S. Frigate	Launched	Disposition
CONSTITUTION	21 Oct 1797	Survivor, still in commission
PRESIDENT	1800	Captured by British in Jan 1815. Broken up in Portsmouth England, 1817
UNITED STATES	10 May 1797	Lost to Confederates Apr 10, 1861. Broken up 1865 in Norfolk, VA.
CONSTELLATION	07 Sep 1797	Broken up in 1853, Norfolk, VA
CONGRESS	1799	Broken up in 1836, Norfolk, VA
CHESAPEAKE	1799	Captured by British, July 1813. Broken up 1820

EDO Forefathers & Notables

- Joshua Humphreys (1751-1838)
  - Principal Constructor of the Navy in 1794
  - Charged (with five other naval architects) with the construction of six war ships for the initial naval needs



- ADM Hyman G. Rickover (1900-1986)
  - Established the modern standards of technical excellence
  - “Father” of the Nuclear Navy
  - Only 4-star Engineering Duty Officer
  - Longest serving Admiral in military history (63 years)



EDO Forefathers & Notables (cont.)

- RADM Wayne E. Meyer (1926-2009)
  - Father of AEGIS
  - Pioneer in systems engineering discipline
  - Developed the functional cornerstones of Detect, Control, Engage
  - Coined the phrase: “Build a Little, Test a Little, Learn a Lot”



- RADM Kate Paige
  - First ASN (RD&A) Chief Engineer
  - First Technical Director of Missile Defense Agency’s Ballistic Missile Defense System
  - Program Director, AEGIS Ballistic Missile Defense



Our Mission and Vision

- ***Mission:*** We design, develop, deliver, and sustain combat-ready naval power for our nation to fight and win—wherever, whenever, and however it chooses

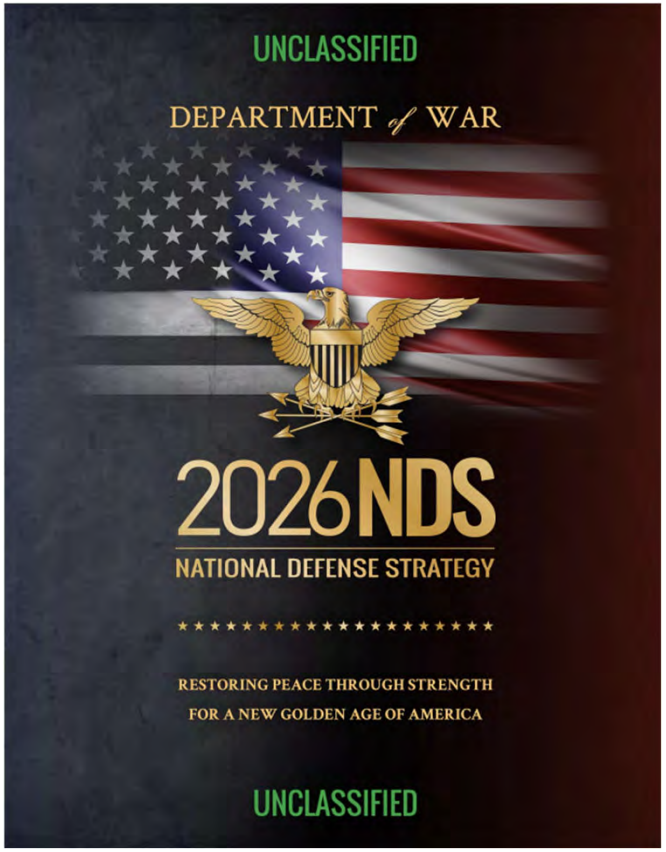
- ***Vision:*** Our Navy wins in all domains by enabling the warfighter through our excellence in engineering, acquisition, maintenance, modernization, repair, and life-cycle ownership, strengthened by forging strategic partnerships at each opportunity

Our Community Role

- To sustain combat readiness and build a Fleet for the future, the Navy requires a cadre of specialized, career Naval Officers who have:
  - Operational experience
  - Technical education and experience
  - Business savvy
    - Navy/DoD processes
    - Industry/business processes
  - Leadership skills
    - Large civilian personnel (CIVPERS) organizations
    - Peers of Industry leaders

Seapower through engineering

2026 National Defense Strategy



- **Keys off of National Security Strategy**
  1. Signed by President Trump, November 2025

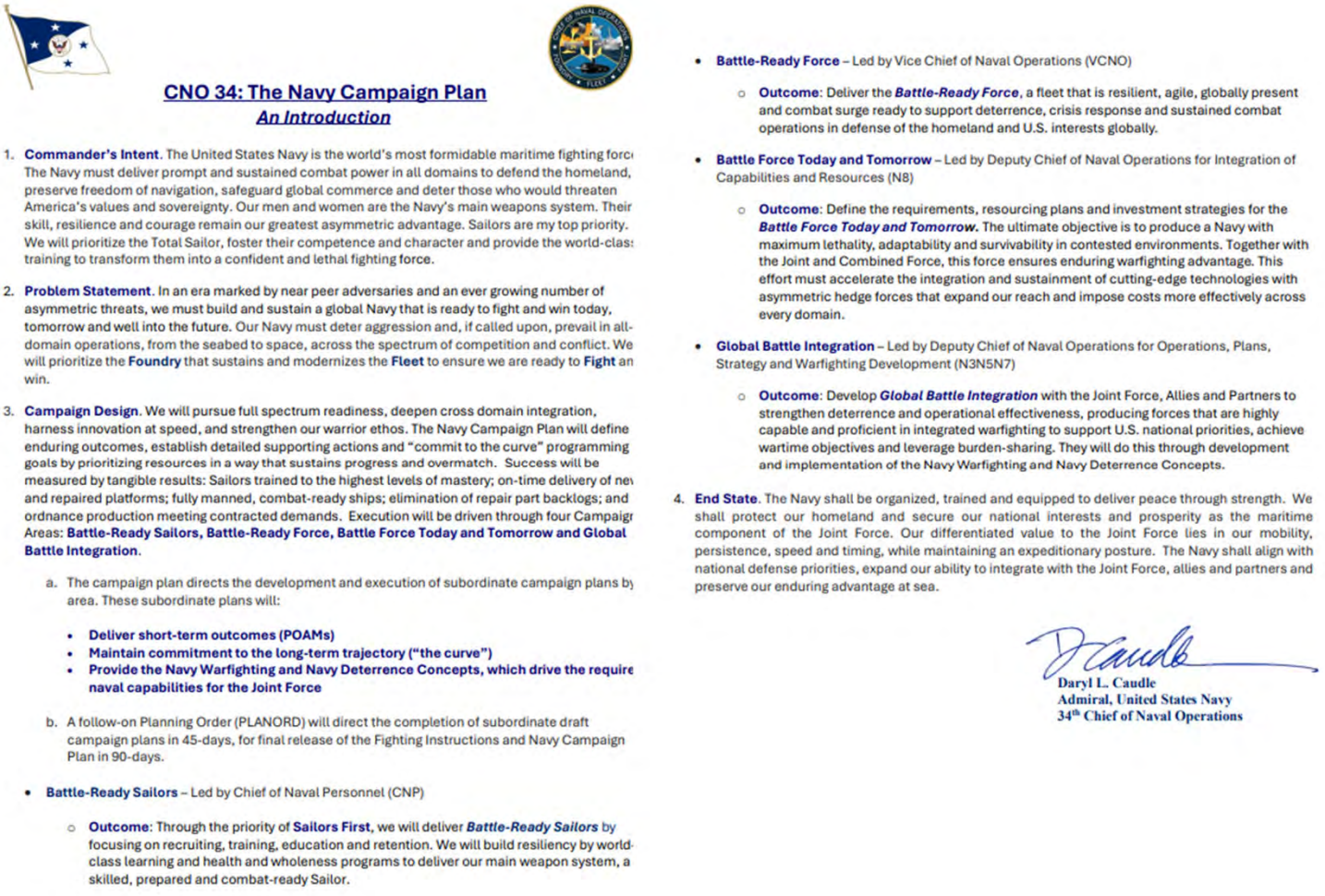
  https://www.whitehouse.gov/wp-content/uploads/2025/12/2025-National-Security-Strategy.pdf
- **Strategic Approach: key lines of effort**
  1. Defend the U.S. Homeland
  2. Deter China in the Indo-Pacific Through Strength, Not Confrontation
  3. Increase Burden-Sharing with U.S. Allies and Partners
  4. Supercharge the U.S. Defense Industrial Base
- **3 ways to advance the priorities**
  1. Integrated deterrence
  2. Campaigning
  3. Building of an enduring advantage

2026 NDS
https://media.defense.gov/2026/Jan/23/2003864773/-1/-1/0/2026-NATIONAL-DEFENSE-STRATEGY.PDF

America First. Peace Through Strength. Common Sense.

CNO Caudle
Foundry – Fleet - Fight

- Commander's Intent
  - C-Notes
- Priorities
  - Foundry
  - Fleet
  - Fight
- Campaign Design
  - Subordinate plans
  - Sailors
    - CNP
  - Battle Ready
    - VCNO
  - Today/Tomorrow
    - N8
  - Global Integration
    - N3N5N7



CNO 34: The Navy Campaign Plan
An Introduction

1. **Commander's Intent**. The United States Navy is the world's most formidable maritime fighting forc The Navy must deliver prompt and sustained combat power in all domains to defend the homeland, preserve freedom of navigation, safeguard global commerce and deter those who would threaten America's values and sovereignty. Our men and women are the Navy's main weapons system. Their skill, resilience and courage remain our greatest asymmetric advantage. Sailors are my top priority. We will prioritize the Total Sailor, foster their competence and character and provide the world-clas training to transform them into a confident and lethal fighting force.

2. **Problem Statement**. In an era marked by near peer adversaries and an ever growing number of asymmetric threats, we must build and sustain a global Navy that is ready to fight and win today, tomorrow and well into the future. Our Navy must deter aggression and, if called upon, prevail in all-domain operations, from the seabed to space, across the spectrum of competition and conflict. We will prioritize the **Foundry** that sustains and modernizes the **Fleet** to ensure we are ready to **Fight** an win.

3. **Campaign Design**. We will pursue full spectrum readiness, deepen cross domain integration, harness innovation at speed, and strengthen our warrior ethos. The Navy Campaign Plan will define enduring outcomes, establish detailed supporting actions and "commit to the curve" programming goals by prioritizing resources in a way that sustains progress and overmatch. Success will be measured by tangible results: Sailors trained to the highest levels of mastery; on-time delivery of ne and repaired platforms; fully manned, combat-ready ships; elimination of repair part backlogs; and ordnance production meeting contracted demands. Execution will be driven through four Campaig Areas: **Battle-Ready Sailors, Battle-Ready Force, Battle Force Today and Tomorrow and Global Battle Integration**.

   a. The campaign plan directs the development and execution of subordinate campaign plans by area. These subordinate plans will:

      - **Deliver short-term outcomes (POAMs)**
      - **Maintain commitment to the long-term trajectory ("the curve")**
      - **Provide the Navy Warfighting and Navy Deterrence Concepts, which drive the require naval capabilities for the Joint Force**

   b. A follow-on Planning Order (PLANORD) will direct the completion of subordinate draft campaign plans in 45-days, for final release of the Fighting Instructions and Navy Campaign Plan in 90-days.

- **Battle-Ready Sailors** – Led by Chief of Naval Personnel (CNP)
  - **Outcome**: Through the priority of **Sailors First**, we will deliver ***Battle-Ready Sailors*** by focusing on recruiting, training, education and retention. We will build resiliency by world-class learning and health and wholeness programs to deliver our main weapon system, a skilled, prepared and combat-ready Sailor.
- **Battle-Ready Force** – Led by Vice Chief of Naval Operations (VCNO)
  - **Outcome**: Deliver the ***Battle-Ready Force***, a fleet that is resilient, agile, globally present and combat surge ready to support deterrence, crisis response and sustained combat operations in defense of the homeland and U.S. interests globally.
- **Battle Force Today and Tomorrow** – Led by Deputy Chief of Naval Operations for Integration of Capabilities and Resources (N8)
  - **Outcome**: Define the requirements, resourcing plans and investment strategies for the ***Battle Force Today and Tomorrow***. The ultimate objective is to produce a Navy with maximum lethality, adaptability and survivability in contested environments. Together with the Joint and Combined Force, this force ensures enduring warfighting advantage. This effort must accelerate the integration and sustainment of cutting-edge technologies with asymmetric hedge forces that expand our reach and impose costs more effectively across every domain.
- **Global Battle Integration** – Led by Deputy Chief of Naval Operations for Operations, Plans, Strategy and Warfighting Development (N3N5N7)
  - **Outcome**: Develop ***Global Battle Integration*** with the Joint Force, Allies and Partners to strengthen deterrence and operational effectiveness, producing forces that are highly capable and proficient in integrated warfighting to support U.S. national priorities, achieve wartime objectives and leverage burden-sharing. They will do this through development and implementation of the Navy Warfighting and Navy Deterrence Concepts.

4. **End State**. The Navy shall be organized, trained and equipped to deliver peace through strength. We shall protect our homeland and secure our national interests and prosperity as the maritime component of the Joint Force. Our differentiated value to the Joint Force lies in our mobility, persistence, speed and timing, while maintaining an expeditionary posture. The Navy shall align with national defense priorities, expand our ability to integrate with the Joint Force, allies and partners and preserve our enduring advantage at sea.

Daryl L. Caudle
Admiral, United States Navy
34th Chief of Naval Operations

Few communities contribute as widely as EDO!

EDO Distinguishing Characteristics

- We are Naval Officers….
  - Operational, warfighter experience
- We have an engineering/technical education
  - Technical master's degree
- We have a business education
  - Warfighting Acquisition University Career training & certification
- We are professionally trained
  - ED Basic Course
  - ED Senior Course
  - ED Captains Seminar
  - Engineering Duty Qualification Program (EDQP)
  - LEAN/Six Sigma

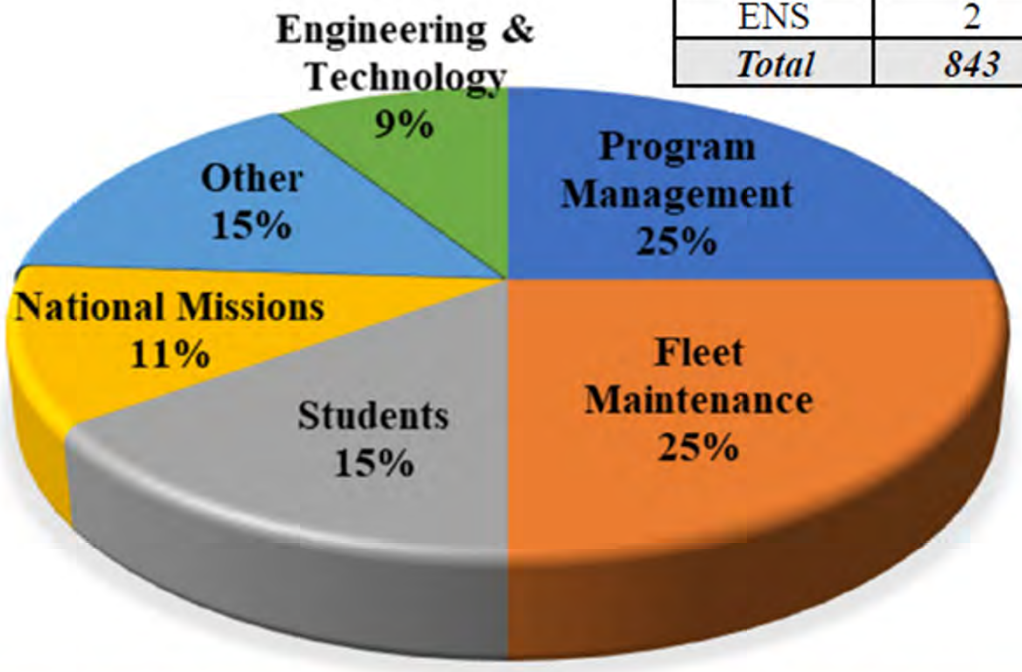
Unique within the acquisition/RL community

EDO Snapshot

- Fleet Maintenance (~25%)
  - Shipyards, RMCs, SRFs, TRFs (19%)
  - TYCOM Staffs (6%)
- Program Management (~25%)
  - PEOs (16%)
  - SUPSHIPs (9%)
- Students (~15%)
- National Missions (~11%)
  - Strategic Systems (6%)
  - Naval Reactors (3%)
  - Space (2%)
- Other (~15%)
  - Staffs (4%)
  - Sea Duty (4%)
  - Instructor Duty, Detailer, Other (3%)
  - TPPH (3%)
- Engineering & Technology (~9%)
  - Warfare/System Centers (5%)
  - HQ Directorates (4%)

	INV	BILLETS
Flag	13	10
CAPT	121	129
CDR	204	208
LCDR	255	272
LT	243	274
LTJG	5	5
ENS	2	0
Total	*843*	*898*

As of 30 Apr 2026



EDs provide Warfare-Qualified, Uniformed Leadership in Primarily Civilian Fields
~850 Officers Lead A Workforce of > 90,000 Civilians

Core Competencies

- Competencies of the Engineering Duty Officer community are common skills that should be understood and mastered by each officer:
  - Leadership
  - Technical acumen
  - Financial management
  - Contracting
  - DoW acquisition
  - Civilian Personnel (CIVPERS) management
  - Battle Damage Assessment and Repair (BDAR)

This course provides the foundation that you will build on the rest of your career

Sub-Specialty Areas and Mentor Groups

- Sub-specialties:
  - Systems Engineering (SE)
  - Warfare Systems (WS)
  - Combat Systems (CS)
  - Hull, Mechanical & Electrical Systems (HM&E)
  - Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR) Systems
  - Ordnance Engineering (ORD)
  - Fleet Maintenance (FM)
  - Program Management (PM)
  - Naval Architecture (NA)
  - Operational Engineering (OpE)
  - Diving & Salvage (D&S)
- Mentor Groups (MG)
  - Industrial Operations (IMG)
  - Surface Maintenance, Modernization & Readiness (SMMR) (formerly RMC)
  - Carriers (CVN CADRE)
  - Combat & Weapons Systems (Cannon Cockers)
  - Information Warfare (IWE)
  - Strategic Systems (SSP)
  - Submarines (UEDO)
  - Surface Ships (SURFPACK)

Sub-specialties and Mentor Groups are loosely correlated

Overview

- EDO School Organization
- Student Introductions
- EDO Community
- EDO School Courses
- CO Expectations

EDO School Courses

- Mission
  - Improve the professional proficiency through training
  - Foster and serve as a focal point for community development of competency and character
- Three primary courses
  - Basic Course
  - Reserve Course
  - Senior Course

Integrated Active-Reserve approach to competency and character development

Basic Course

- Purpose
  - Provide the fundamental skills and practical knowledge necessary to assume responsibilities for management of engineering tasks associated with
    - Research and Development (R&D)
    - Design
    - Acquisition
    - Construction
    - Maintenance and modernization of ships and systems
  - Transition from Unrestricted Line to Restricted Line roles and responsibilities
  - Begin developing personal and professional networks
  - Familiarize with mentoring/career planning process/tools
  - Set foundation for self-awareness/self-improvement journey

Prepare you for an effective and rewarding initial EDO tour

Basic Course (cont.)

- Highlights
  - 5-week course length
    - 2 weeks for RC
  - Mix of staff and visiting instructors
  - ACQ 1010, 2020, 2030 and PMT 2570V Equivalency
    - 4 of 16 Warfighting University PM (Practitioner) training requirements
    - Warfighting Acquisition University - https://www.waru.edu/
    - Active Component only
  - MBTI/CPI-260 assessments
    - Self-awareness
    - Team dynamics
  - Counseling
    - One-on-one detailer
    - Mentor Group

Basic Course Focus

- Obtain knowledge and awareness of:
  - Plans, programs, policies, and procedures used to accomplish life-cycle engineering of naval ships and ship systems
  - Organizations where EDOs have major responsibilities
    - Practical aspects of EDO career planning
  - Industrial technologies, processes, and procedures used in the production and maintenance of naval ships and ship systems
  - Planning, Programming, Budgeting, and Execution (PPBE)
  - Working with, for, and leading civilians
  - Current and upcoming innovation opportunities and rapid capabilities to support improving outcomes

Basic Course Focus (cont.)

- Obtain knowledge and awareness of:
  - Program funding, field activity financial management, and other financial topics
    - Shipyards, Supervisors of Shipbuilding, Regional Maintenance Centers, and Warfare Centers
  - Acquisition and contracting
  - Leadership/self-assessment/Warrior Toughness
- Practical application of:
  - Teambuilding
  - Leadership
  - Briefing techniques
  - Integrated case studies and scenarios
  - Critical thinking

Basic Course

- ADMIN (1.x.x courses)
- Command Structures (2.x.x courses)
- Acquisition (3.x.x courses)
  - ACQ 1010, 2020, 2030, PMT 2570V
- Fleet Maintenance (4.x.x courses)
- Professional Development/Leadership (5.x.x courses)
- Civilian Personnel Management (6.x.x courses)

Reserve Course

- ADMIN (1.x.x courses)
- Command Structures (2.x.x courses)
- Acquisition (3.x.x courses)
  - Not all of the course but a good dose
- Fleet Maintenance (4.x.x courses)
  - A few of the courses
- Professional Development/Leadership (5.x.x courses)
  - Presentations techniques and FITREPS

Main difference is less acquisition (WarU) and fleet maintenance coursework

Senior Course

- Purpose: Prepare newly selected commanders for increased responsibility associated with…
  - Research and Development
  - Design
  - Acquisition
  - Construction
  - Maintenance and modernization of ships and ship systems as senior Naval Officer
  - ***Organizational leadership***

Same functions, different roles and responsibilities

Senior Course – Major Topics

- FLAG presentations:
  - Current Navy/EDO leadership issues and challenges
- CAPT presentations:
  - Leadership issues and challenges from EDO core areas
- EDO Community Plans and Policy
  - Career and professional development
- CPI 260 – Leadership development/self-awareness
- OPM 360 – Leadership development
- Media and legal training
- Point papers and two-minute elevator speeches
- Reserve EDO Program

Community and leadership focus

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Basic Ground Rules

- ***Non-Attribution Policy***
  - Applies to speakers, staff, and fellow students
  - See EDOSCOL Notice
- Academic environment
  - Do your homework
  - You will demonstrate your knowledge through tests, presentations, papers, and in class discussions
  - You will be called on in class to answer questions on the material
- Liberty
  - San Diego to Monterey
- Leave:
  - Generally, not authorized for EDO Basic Course

Basic Ground Rules (cont.)

- Cell phones are ***not*** allowed in the classroom
  - Receptacles in CD2's office
- Eating in the classroom is a privilege – don't lose it!
  - Use good judgement – it cannot be a distraction to the instructor/speaker or students
  - Ensure the classroom is clean at the end of each day
- Reserve Course ends by noon on Friday of week 2
- Basic Course ends by noon on Friday of week 5
- Early departures will not be approved

Expectations

- Core Values (Honor, Courage, Commitment)
  - Respect each other, staff and guests
  - Support your CD, class leader, and each other
- Ethical behavior
  - Zero tolerance for cheating
- Be respectful to presenters
  - No multi-tasking on computers during presentations
- Practice Operational Risk Management (ORM)
  - TRiPS, high risk athletic endeavors
  - Responsible use of alcohol
- Sexual harassment/assault
  - Zero tolerance
  - Practice bystander intervention
  - EDO School SAPR UVA is LCDR Girouard
- Take care of yourself (and your family)

Summary

- Be as excited to be here as we are to have you here
- Expectations are high but achievable
- Always act in a highly professional and ethical manner
- Take the non-attribution policy seriously
- Build your networks
- Take advantage of the mentoring and counseling opportunities

Welcome to the EDO community and your school!